CIS 121 Introduction to Programming
Sequential code and the if statement

1. Allow a user to enter a quantity of an item. If the quantity is greater than or equal to 1000, the unit price should be $3.00. For quantities under 1000 the unit price is $5.00. Compute extended price to be quantity x unit price. Compute tax to be 7% of the extended price. The total is computed as extended price plus the tax. Display the quantity, unit price, extended price, tax and total.

Input	Process	Output
• Quantity of items	• Assign a variable to the input for the Quantity • Declare Unit Price to be $3 if quantity is greater than or equal to 1000 • Otherwise, set Unit Price to $5 • Calculate the Extended Price by finding the product of Quantity and Unit Pirce • Calculate Tax by finding the product of Extended Price and 0.07 • Calculate Total by finding the sum of Extended Price and Tax • Display Quantity, Unit Price, Extended Price, Tax, and Total	• Quantity of items • Unit Price • Extended Price • Tax • Total

2. The program asks the user for an item and quantity. Determine the unit price of the item based on the chart below. Compute the extended price to be quantity x unit price. Display the item, unit price and extended price.

Item	Unit Price
A	$10.00
B	$20.00

Input	Process	Output
• Item • Quantity	• Assign a variable to the input for Item • Set Unit Price to $10 if Item A was selected • Set Unit Price to $20 if Item B was Selected • Assign a variable to the input for the Quantity • Calculate the extended price by finding the product of Quantity and the item's Unit Price • Display Item, Unit Price, and Extended Price	• Item • Unit Price • Extended Price

3. Enter the number of books to order and cost per book. If the order total is over $50.00 shipping is free. If the order total is $50.00 or under charge $25 shipping. Display the order total and shipping charge (note 0 should display for a free shipping charge).

Input	Process	Output
• # of Books • Cost per Book	• Assign a variable to the input for # of Books • Assign a variable to the input for Cost per Book • Calculate the Order Total by finding product of # of Books and Cost per Book • Declare Shipping Charge to be $25 if Order Total is less than or equal to 50 • Otherwise, set the Shipping Charge to be $0 • Display Order Total and Shipping Charge	• Order Total • Shipping Charge

4. The warrantee of an appliance depends on the cost of the appliance. For appliances over $1,000 the warrantee cost is 10% of the price. For appliances $1,000 or less the warrantee cost is 5% of the price. The user will enter the name and cost of an appliance. Display name and cost of appliance, the cost of the warrantee and the total (cost of the appliance + warranty).

<table>
<tr><th>Input</th><th>Process</th><th>Output</th></tr>
<tr>
<td>• Name of Appliance
• Cost of Appliance</td>
<td>• Assign a variable to the input for Name of Appliance
• Assign a variable to the input for Cost of Appliance
• Declare Cost of Warrantee to be 10% (.10) if Cost of Appliance is greater than $1000
• Otherwise, set Cost of Warrantee to be 5% (.05)
• Calculate Total by finding the sum of Cost of Appliance and Cost of Warrantee
• Display Name of Appliance, Cost of Appliance, Cost of Warrantee, and Total</td>
<td>• Name of Appliance
• Cost of Appliance
• Cost of Warrantee
• Total</td>
</tr>
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5. Enter the user's last name, number of dependents and gross income. Compute adjusted gross income to be gross income minus dependents times $12000. Next determine an income tax rate. Adjusted gross incomes over $50,000 have a tax rate of 20%. Adjusted gross incomes $50,000 or under have a tax rate of 10%.

   Once you determine the tax rate, compute income tax to be adjusted gross income times tax rate. If the income tax is less than 0, set the income tax to $100.

   Display last name, gross income, number of dependents, adjusted gross income, and income tax.

Input	Process	Output
• User's Last Name • # of Dependents • User's Gross Income	• Assign a variable to the input for User's Last Name • Assign a variable to the input for # of Dependents • Assign a variable to the input for Gross Income • Calculate Adjusted Gross Income by finding the difference of cost Gross Income and # of Dependents times $12000 • Declare Tax Rate to be 20% (.20) if Adjusted Gross Income is greater than $50,000 • Otherwise, set Tax Rate to be 10% (.10) • Calculate Income Tax by finding the product of Adjusted Gross Income and Tax Rate • Declare Income Tax to be $100 if Income Tax is less than $0, then Display User's Last Name, # of Dependents, User's Gross Income, Adjusted Gross Income, and Income Tax • Otherwise, Display User's Last Name, # of Dependents, User's Gross Income, Adjusted Gross Income, and Income Tax	• User's Last Name • # of Dependents • User's Gross Income • Adjusted Gross Income • Income Tax

Example Programs - In class

1. Input a student's last name and GPA, display the last name, GPA and a message of "Well Done" when the GPA is greater than or equal to 4.0. When the GPA is less than 4.0 display a message "Good Effort".

2. Allow a user to enter the meal cost. For meals over $25 compute a tip to be 20% of the meal. For meals $25 or under, give a $3.00 tip. Compute the total to be cost of meal plus the tip. Display the meal cost, tip and total.

3. The user enters the type of gasoline of either "regular" or "premium". The user will also enter the gallons of gas purchased. Premium gas costs $2.75 and regular costs $2.25 per gallon. Compute the total to be gallons purchased times cost per gallon. Display type of gas, gallons purchased, cost per gallon and total.

4. Allow a user to enter the base, height and hypotenuse of a triangle. Display a message, "Is a Right Angle" when the hypotenuse is equal to the square root of the sum of the squares of the base and height. Display "not a Right Angle" if this is not the case.

5. Allow the user to enter two numbers. If the first number is greater than the second number display a message "Is Greater" otherwise display a message "Is Not Greater".